Name of Student: Rikisha Bhattacharya	**PRN: 25070122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 4	
TITLE: Wrapper Classes in Java	

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

1.Create a program to convert student marks from String format to Integer and calculate total marks

Programme

```java
// Java Program to Demonstrate Wrapper Classes
public class Assignment4 {

    Run | Debug
    public static void main(String[] args) {

        // Marks in String format
        String m1 = "85";
        String m2 = "90";
        String m3 = "78";

        // Convert String to Integer
        int mark1 = Integer.parseInt(m1);
        int mark2 = Integer.parseInt(m2);
        int mark3 = Integer.parseInt(m3);

        // Calculate total
        int total = mark1 + mark2 + mark3;

        System.out.println(x: "Student Marks");
        System.out.println("Mark 1 = " + mark1);
        System.out.println("Mark 2 = " + mark2);
        System.out.println("Mark 3 = " + mark3);
        System.out.println("Total Marks = " + total);
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58661' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment4'
Student Marks
Mark 1 = 85
Mark 2 = 90
Mark 3 = 78
Total Marks = 253
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Program

```java
// Java Program to Demonstrate Wrapper Classes
import java.util.Scanner;

public class Assignment4 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee ID: ");
        String empId = sc.nextLine();

        System.out.print("Enter Basic Salary: ");
        String basic = sc.nextLine();

        System.out.print("Enter Bonus: ");
        String bonus = sc.nextLine();

        // Convert String to Wrapper Objects
        Integer basicSalary = Integer.valueOf(basic);
        Integer bonusAmount = Integer.valueOf(bonus);

        // Validation
        if (basicSalary < 0 || bonusAmount < 0) {
            System.out.println("Invalid salary or bonus.");
        } else {
            Integer netSalary = basicSalary + bonusAmount;

            System.out.println("\nEmployee ID: " + empId);
            System.out.println("Basic Salary: " + basicSalary);
            System.out.println("Bonus: " + bonusAmount);
            System.out.println("Net Salary: " + netSalary);
        }

        sc.close();
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58322' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment4'
Enter Employee ID: 34567
Enter Basic Salary: 70000
Enter Bonus: 5000

Employee ID: 34567
Basic Salary: 70000
Bonus: 5000
Net Salary: 75000
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

The programs successfully demonstrated the use of Java wrapper classes for converting primitive data types and strings into objects and vice versa. They also showed how wrapper classes can be used for data conversion, validation, and basic arithmetic operations, making Java programs more flexible and efficient when working with object-oriented features and collections.